Computer Networks (CS-241)

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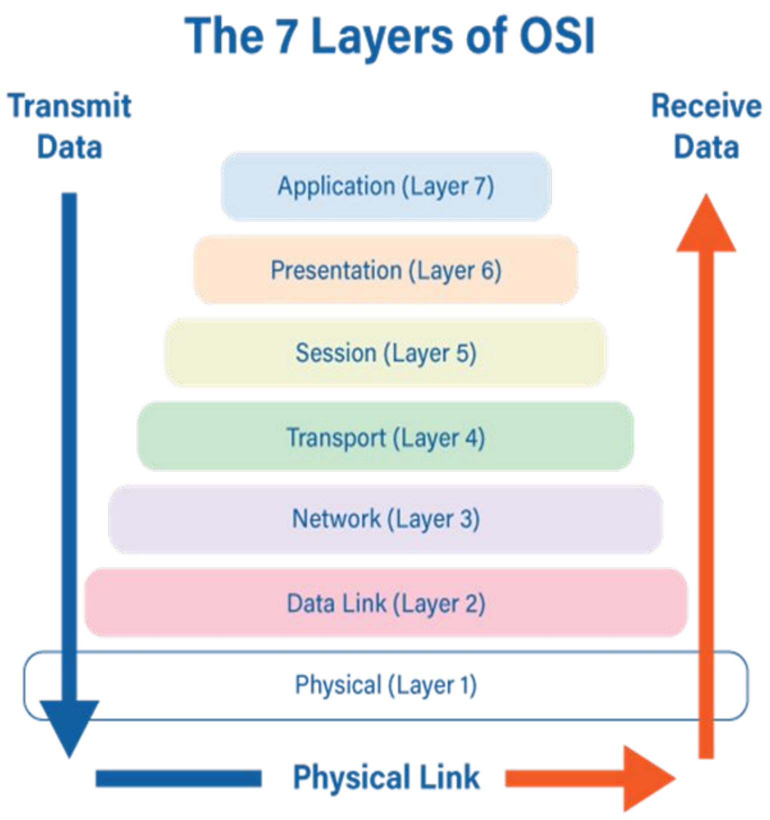


Adapted From:

Computer Networking: A Top-Down Approach

8th edition: Jim Kurose, Keith Ross Pearson, 2020

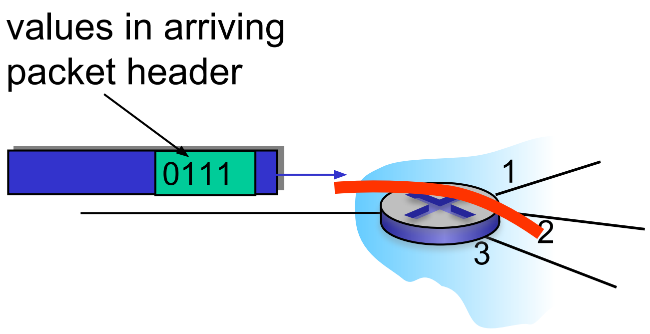
Network layer: "data plane" roadmap

- Network layer: overview
  - data plane
  - control plane
  - Forwarding & routing
- What's inside a router
  - input ports, switching, output ports
  - buffer management, scheduling
- IP: the Internet Protocol
  - datagram format
  - addressing
  - network address translation
  - IPv6

Network layer: data plane, control plane

Data plane:

- *local*, per-router function
- determines how datagram arriving on router input port is forwarded to router output port



Control plane

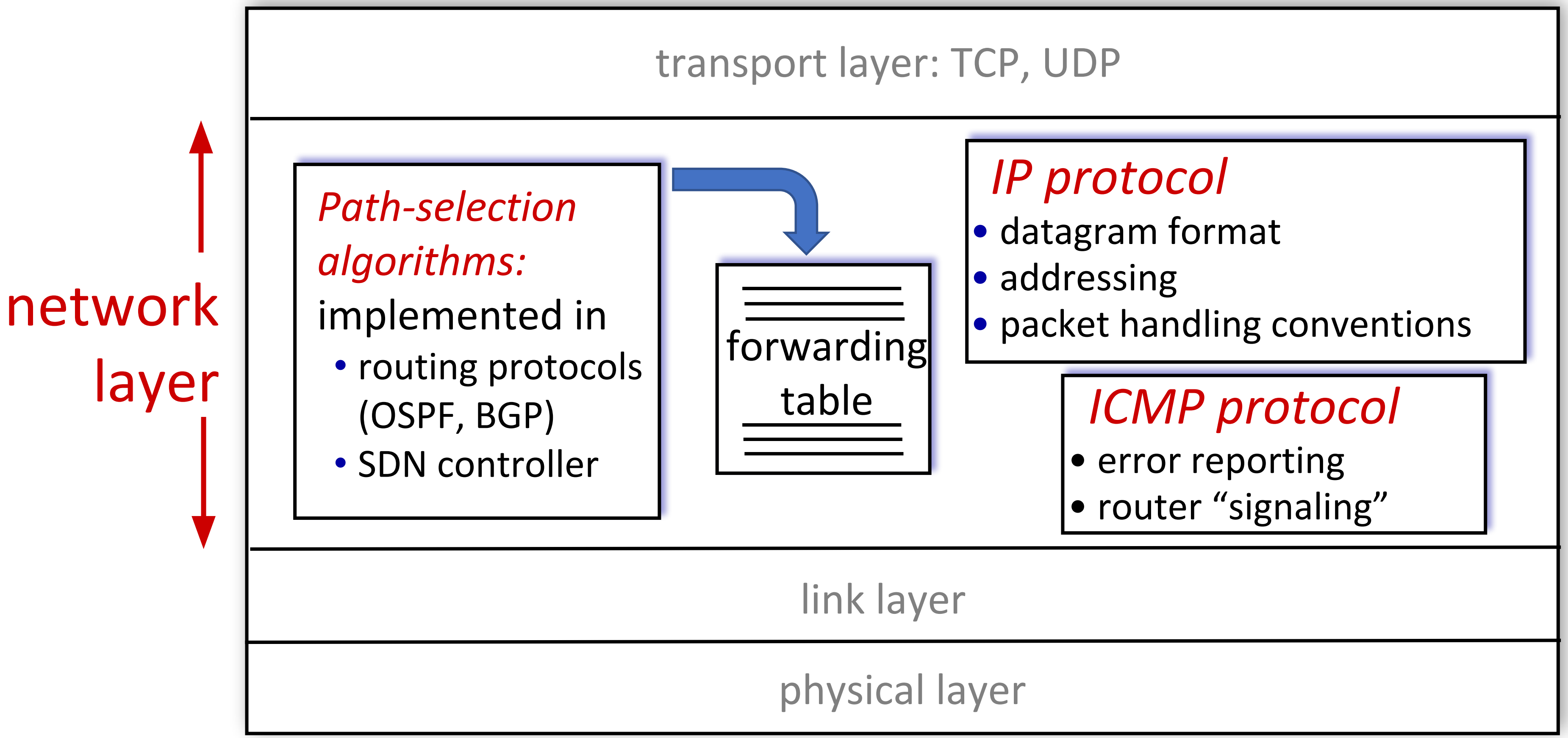
- *network-wide* logic
- determines how datagram is routed among routers along end-end path from source host to destination host
- two control-plane approaches:
  - *traditional routing algorithms:* implemented in routers
  - *software-defined networking (SDN)*: implemented in (remote) servers

Comparison Table

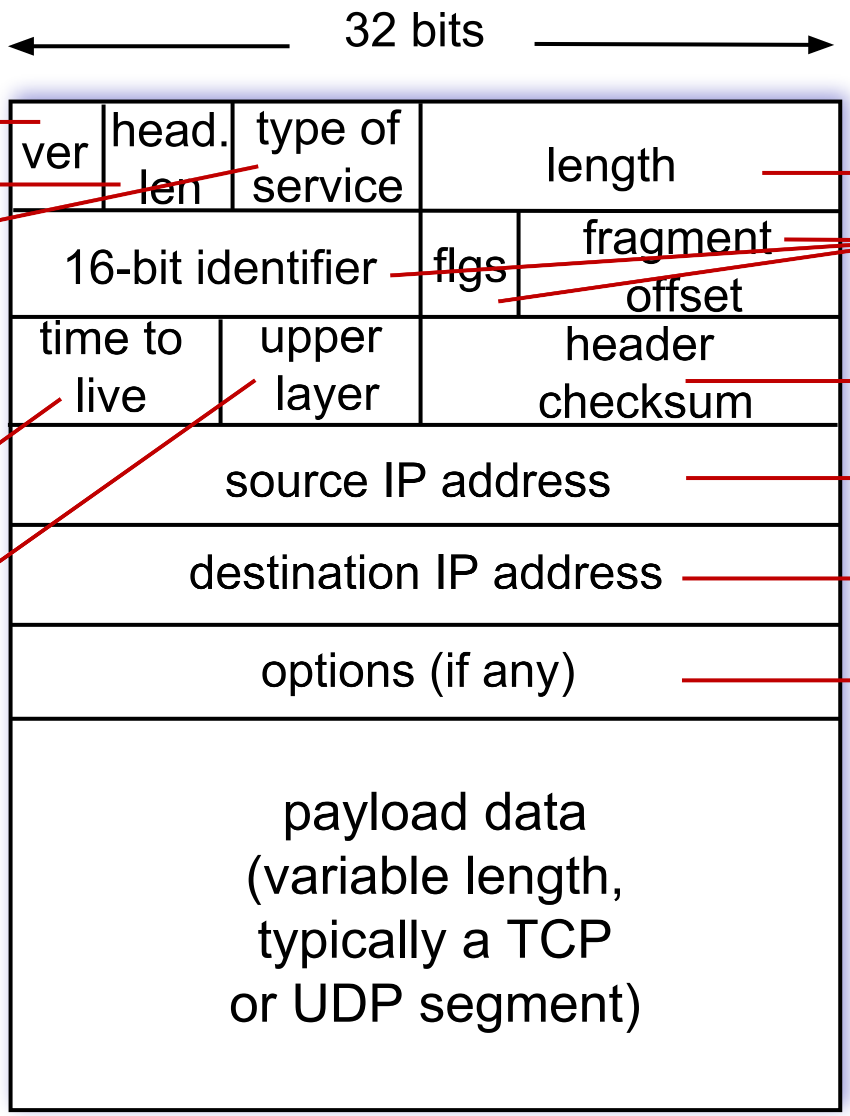
Data Plane	Control Plane
Forwards packets	Decides routes
Fast operation	Slower operation
Uses forwarding table	Builds forwarding table
Packet-by-packet work	Network-wide decisions

Network Layer: Internet

host, router network layer functions:



IP Datagram format/header



IP protocol version number

header length(bytes)

“type” of service means type of data
- HTTP
- FTP

TTL: limits the lifetime of a packet in a network When TTL expire indicate “**Timeout**”

upper layer(Transport Layer) protocol (e.g., TCP or UDP)

overhead
- 20 bytes of TCP
- 20 bytes of IP
- = 40 bytes + app layer overhead for TCP+IP

total datagram length (16-bytes)

fragmentation/ reassembly

header checksum

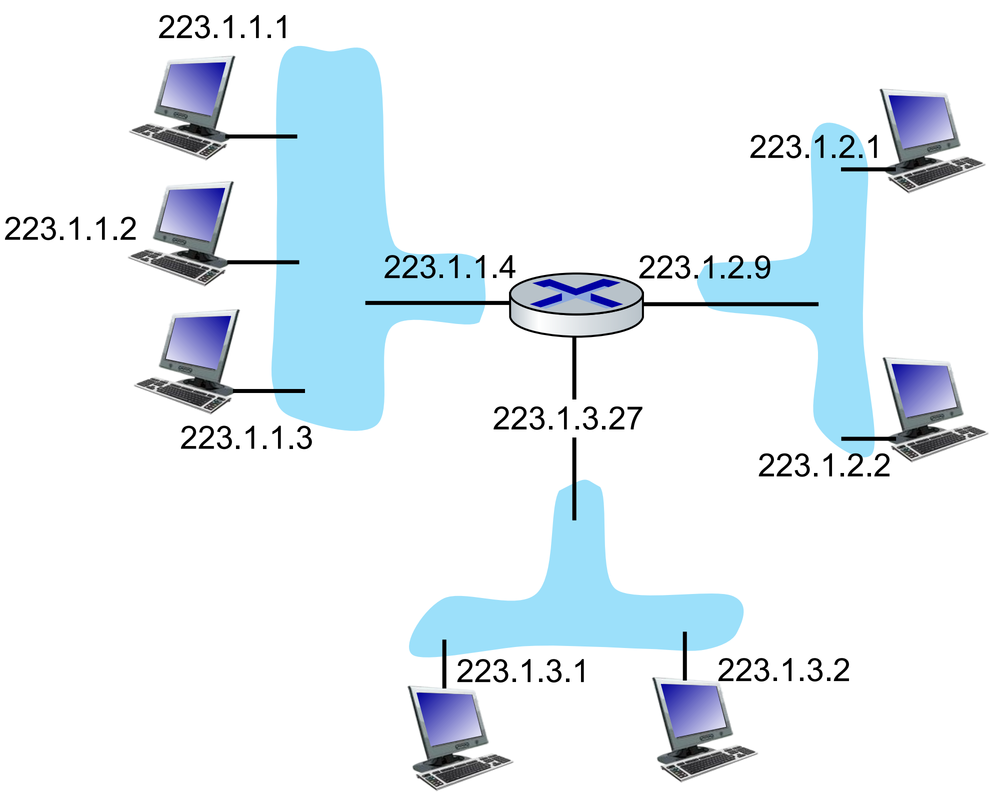
32-bit source IP address

Maximum length: 64K bytes
Typically: 1500 bytes or less

e.g., timestamp, record the route taken

IP addressing: introduction

- IP address: An IP address is a 32-bit number (in IPv4). It uniquely identifies a host or a router interface on the network
- **Example:** 192.168.1.1
- interface: An interface is the connection point between a device (host or router) and a physical network link.
- IP addresses are assigned to interfaces, not to the whole device
- host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)



dotted-decimal IP address notation:

$$223.1.1.1 = \underbrace{11011111}_{223}\ \underbrace{00000001}_{1}\ \underbrace{00000001}_{1}\ \underbrace{00000001}_{1}$$

IP addressing

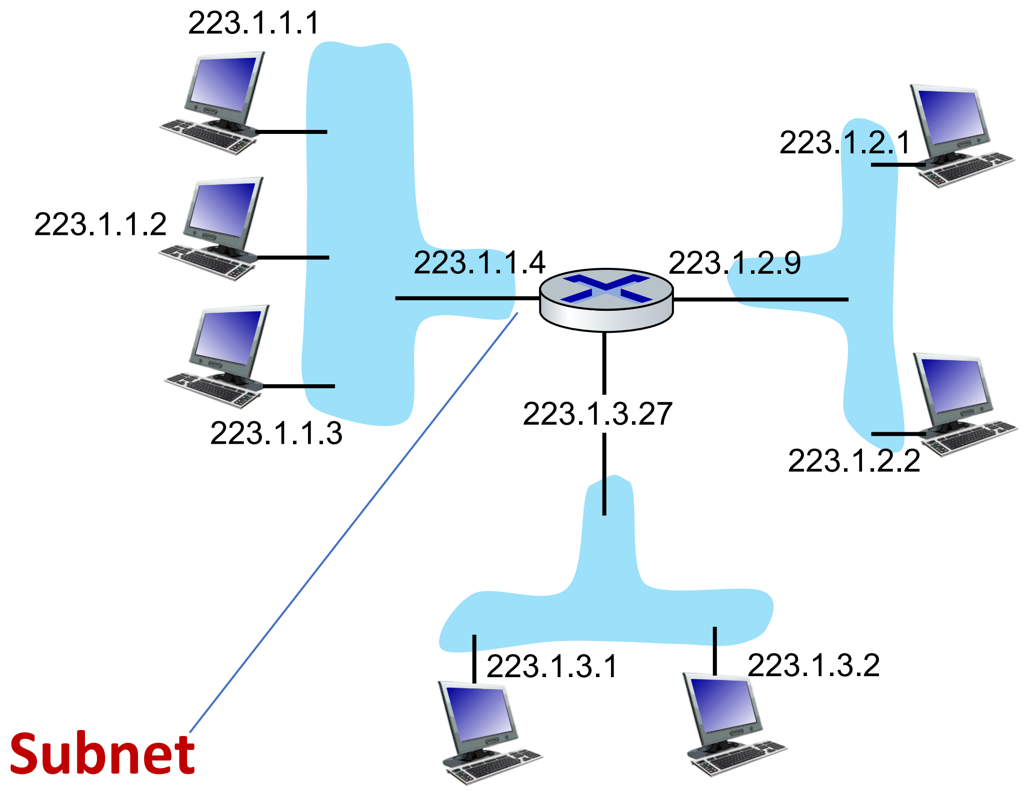
Example: 192.168.1.1

Indicate Class Network part Subnet-Network Host

Class	First Octet Range	Use
A	0 – 127	Large networks
B	128 – 191	Medium networks
C	192 – 223	Small networks
D	224 – 239	Multicast
E	240 – 255	Experimental

IP addressing: introduction

- This is a **small IP network** connected via a **router**.
- There are **3 subnets**:
  - **Subnet A (223.1.1.x)** – top-left
  - **Subnet B (223.1.2.x)** – top-right
  - **Subnet C (223.1.3.x)** – bottom
- The **router** connects all three subnets.
- Each device (PC) has a **unique IP address**.
- First **3** bytes **(223.1.1)** → **Network ID**
  Last byte (1) → **Host ID**



dotted-decimal IP address notation:

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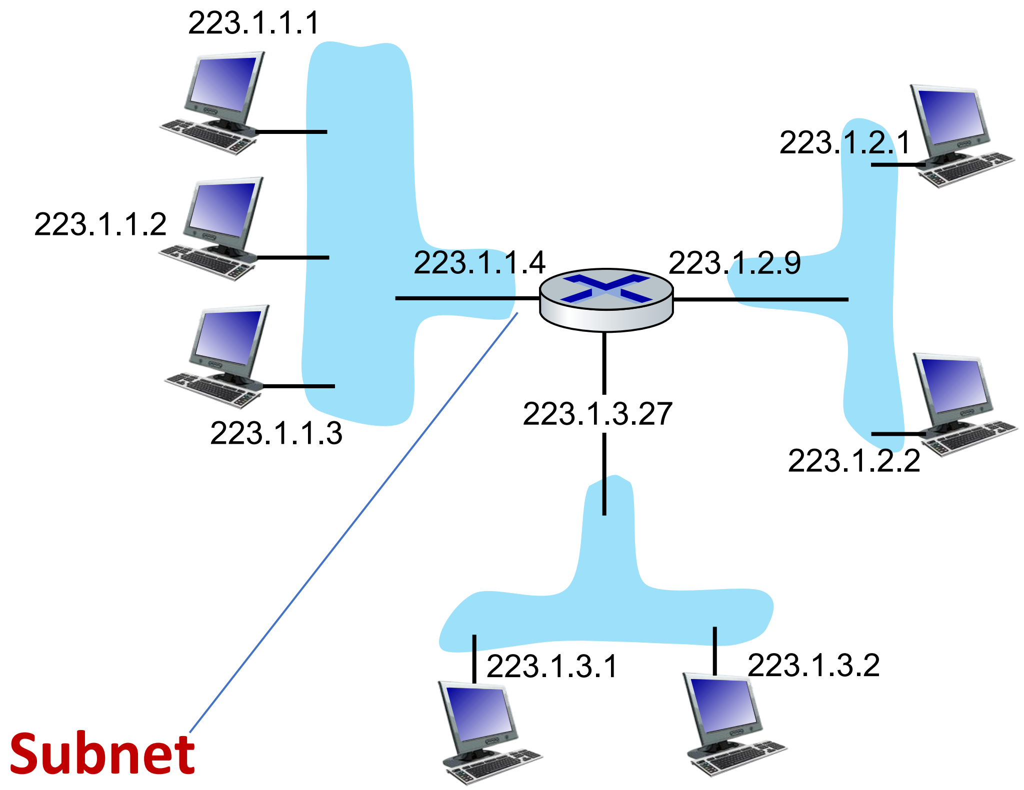
IP addressing: introduction

1. **Sub-netting**
   1. Each group of connected PCs is a **subnet**
   2. Router interfaces are part of each subnet to **route traffic** between subnets
2. **Routing**
   1. When a PC wants to send data to a PC in **another subnet**, it sends it to the **router interface**.
   2. Router checks the **destination subnet** and forwards the packet accordingly.



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IP addressing: introduction

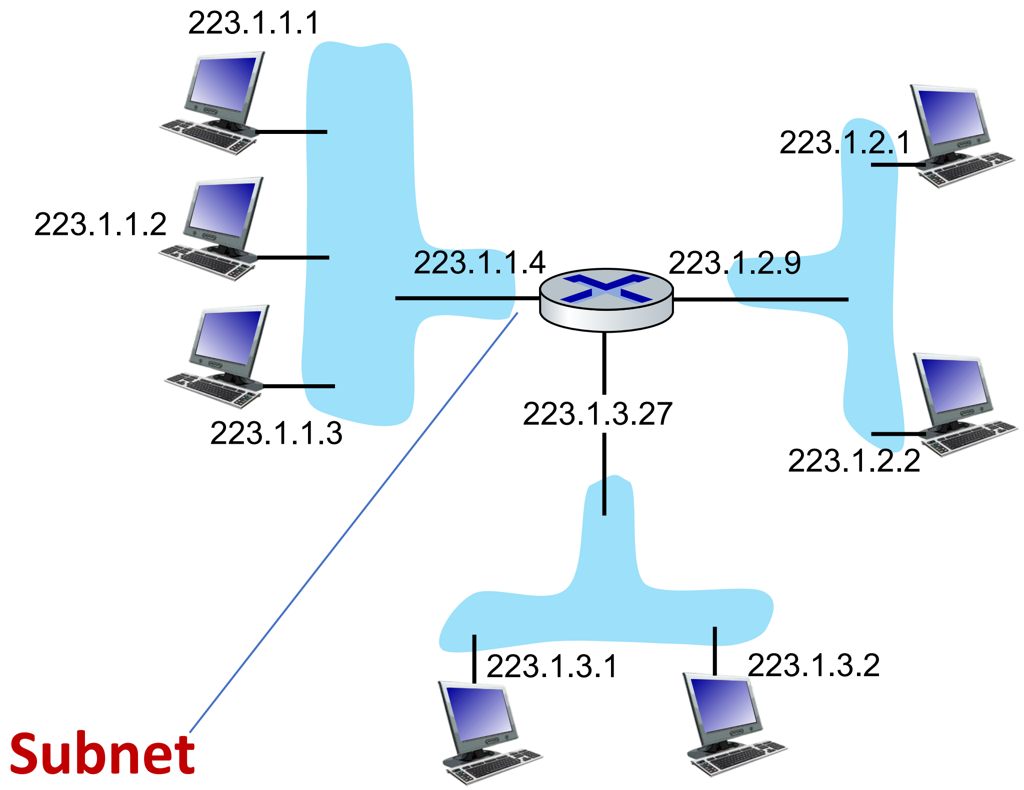
Example Communication

Case 1: Same subnet

- PC 223.1.1.1 → PC 223.1.1.3
- Packet goes **directly**, no router needed

Case 2: Different subnet

- PC 223.1.1.1 → PC 223.1.2.2
- Packet goes to **router interface 223.1.1.4**, then router forwards it to **223.1.2.9**, finally to **PC 223.1.2.2**



dotted-decimal IP address notation:

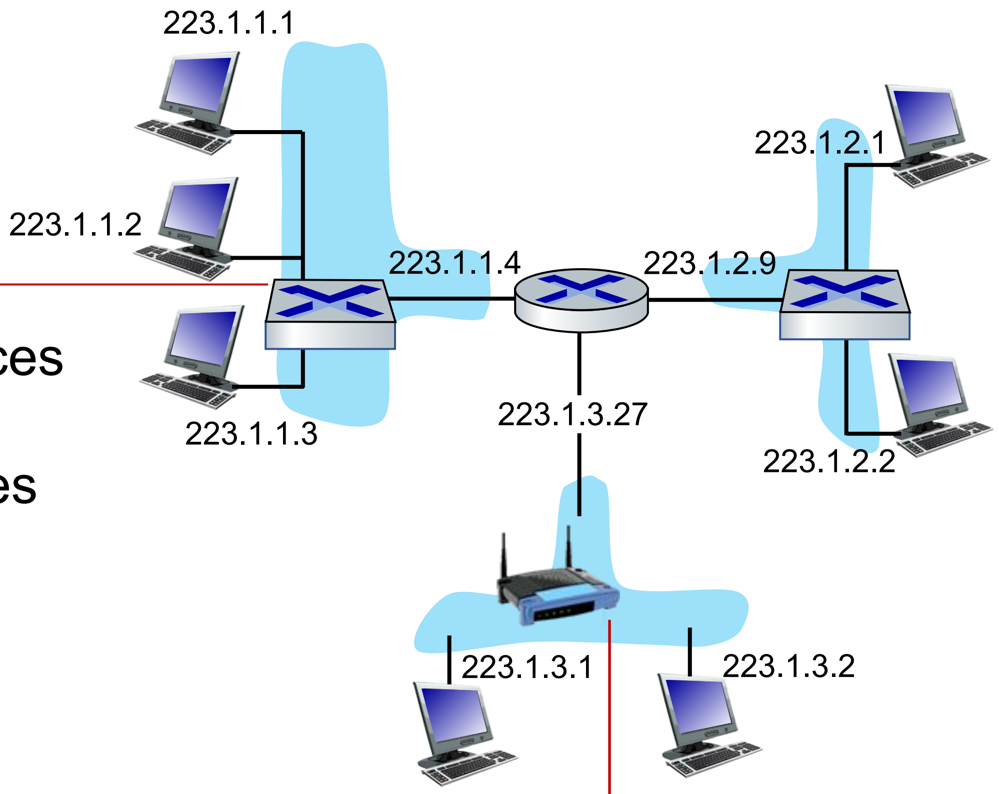
$$223.1.1.1 = \underbrace{11011111}_{223}\ \underbrace{00000001}_{1}\ \underbrace{00000001}_{1}\ \underbrace{00000001}_{1}$$

IP addressing: introduction

Q: how are interfaces actually connected?

A: we'll learn about that in chapters 6, 7

A: wired Ethernet interfaces connected by Ethernet switches

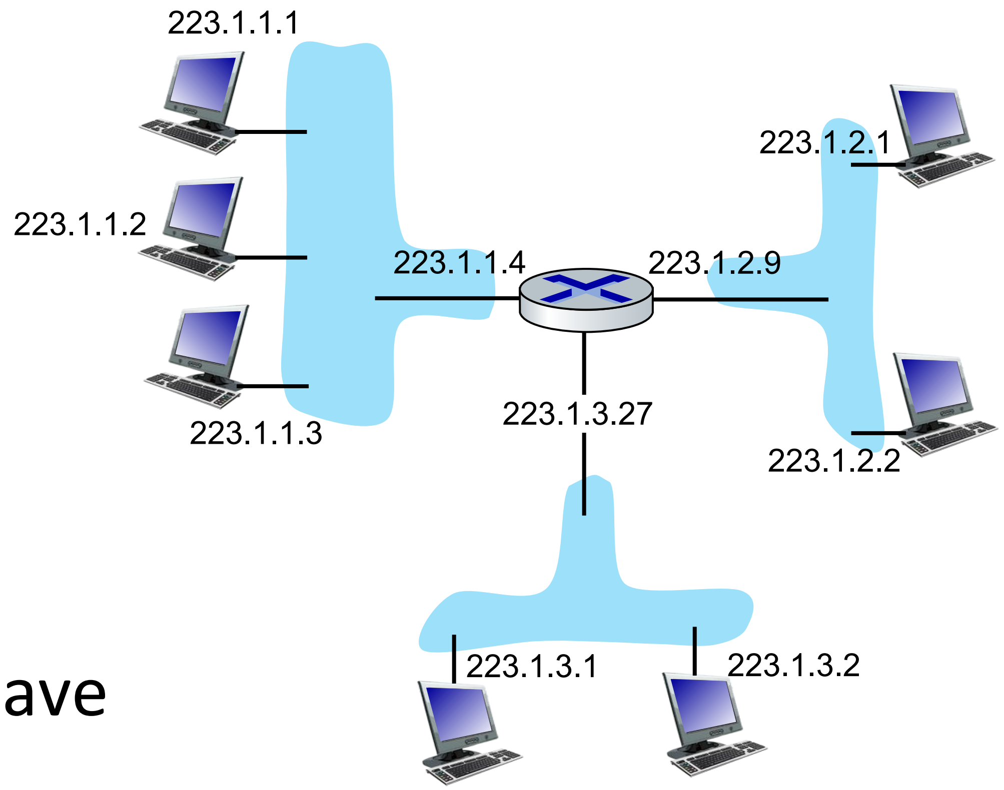


Laptop (Wi-Fi interface)
↓
Wi-Fi Base Station (AP)
↓
Network / Internet

A: wireless WiFi interfaces connected by WiFi base station(access point)

Subnets

- *What's a subnet ?*
  - A **subnet (subnetwork)** is a **small part of a larger network**.
  - It divides one big network into **smaller networks**.
- IP addresses have structure:
  - subnet part: devices in same subnet have common high order bits(left side bits)
  - host part: remaining low order bits

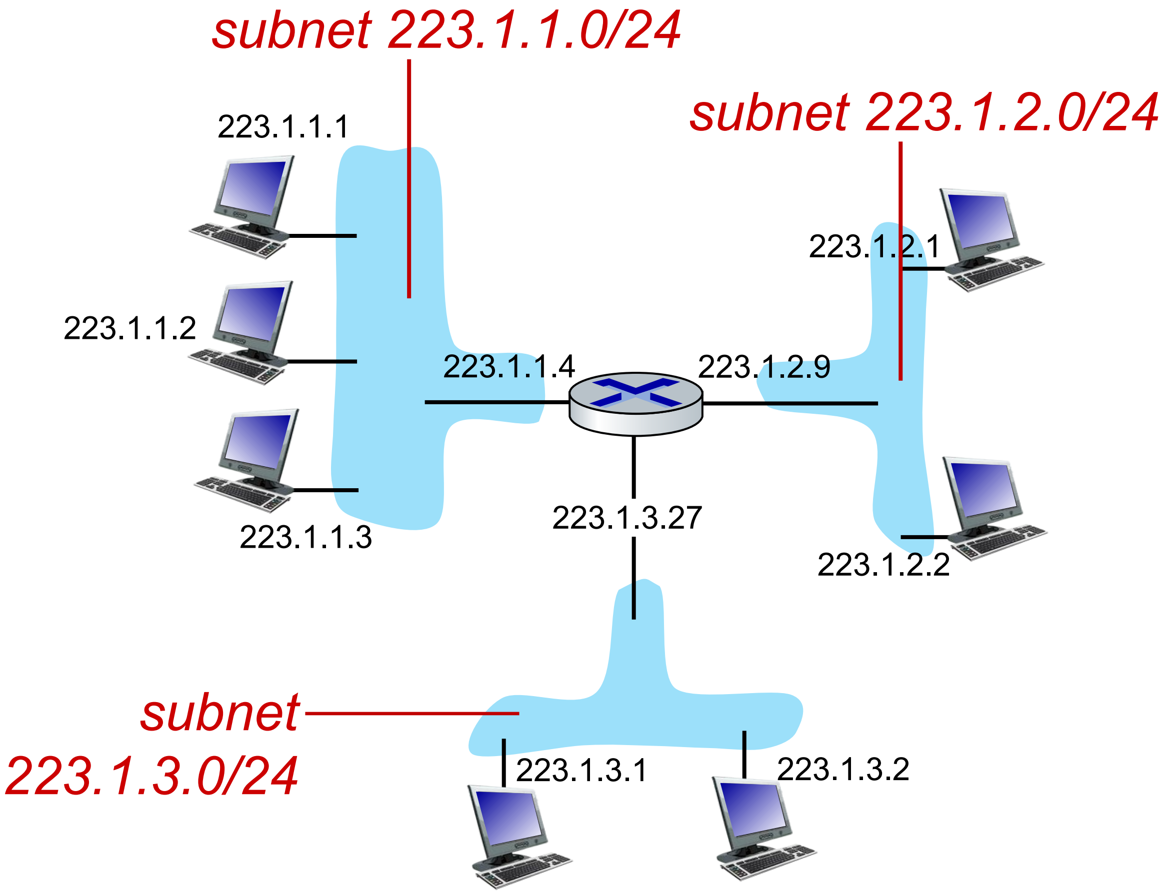


network consisting of 3 subnets

Subnets

Recipe for defining subnets:

- To define subnets, detach each interface from hosts or routers; each resulting isolated network forms a subnet.
- Imagine you cut the connection between networks (only in your mind):
- Devices that are still connected together form one isolated network.
- They cannot reach other devices without a router
- **That isolated group = subnet**



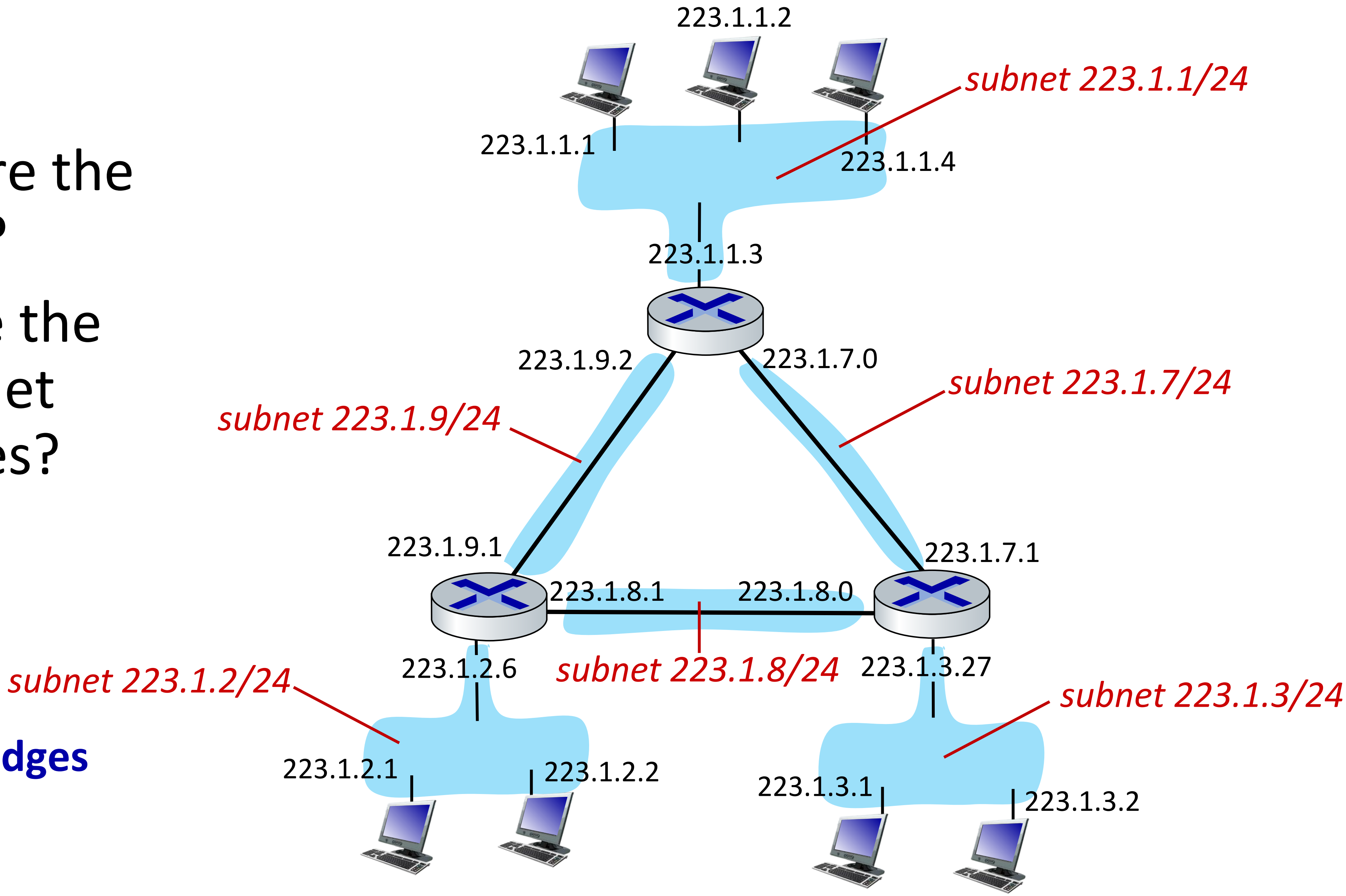
subnet mask: /24

(high-order 24 bits: subnet part of IP address)

Most Common Subnet: 255.255.255.0

Subnets

- where are the subnets?
- what are the /24 subnet addresses?



- **Routers act like bridges between Subnets.**

IP addressing: CIDR

CIDR: Classless InterDomain Routing (pronounced "cider")

- subnet portion of address of arbitrary length
- address format: a.b.c.d/x, where x is # bits in subnet portion of address

subnet part | host part

11001000 00010111 00010000 00000000

200.23.16.0/23

IP addresses: how to get one?

That's actually two questions:

1. Q: How does a *host* get IP address within its network (host part of address)?
   - A. By DHCP and By Network Administrator
2. Q: How does a *network* get IP address for itself (network part of address)
   - A. network gets its IP address from a higher authority.

Example: Company networks and home networks get IP blocks from **ISP.**

How does *host* get IP address?

A host can get an IP address in two ways:

1. **hard-coded** by system admin in config file (e.g., /etc/rc.config in UNIX)
2. DHCP: Dynamic Host Configuration Protocol: dynamically get address from as server
   - "plug-and-play"

DHCP: Dynamic Host Configuration Protocol

goal: host *dynamically* obtains IP address from network server when it "joins" network

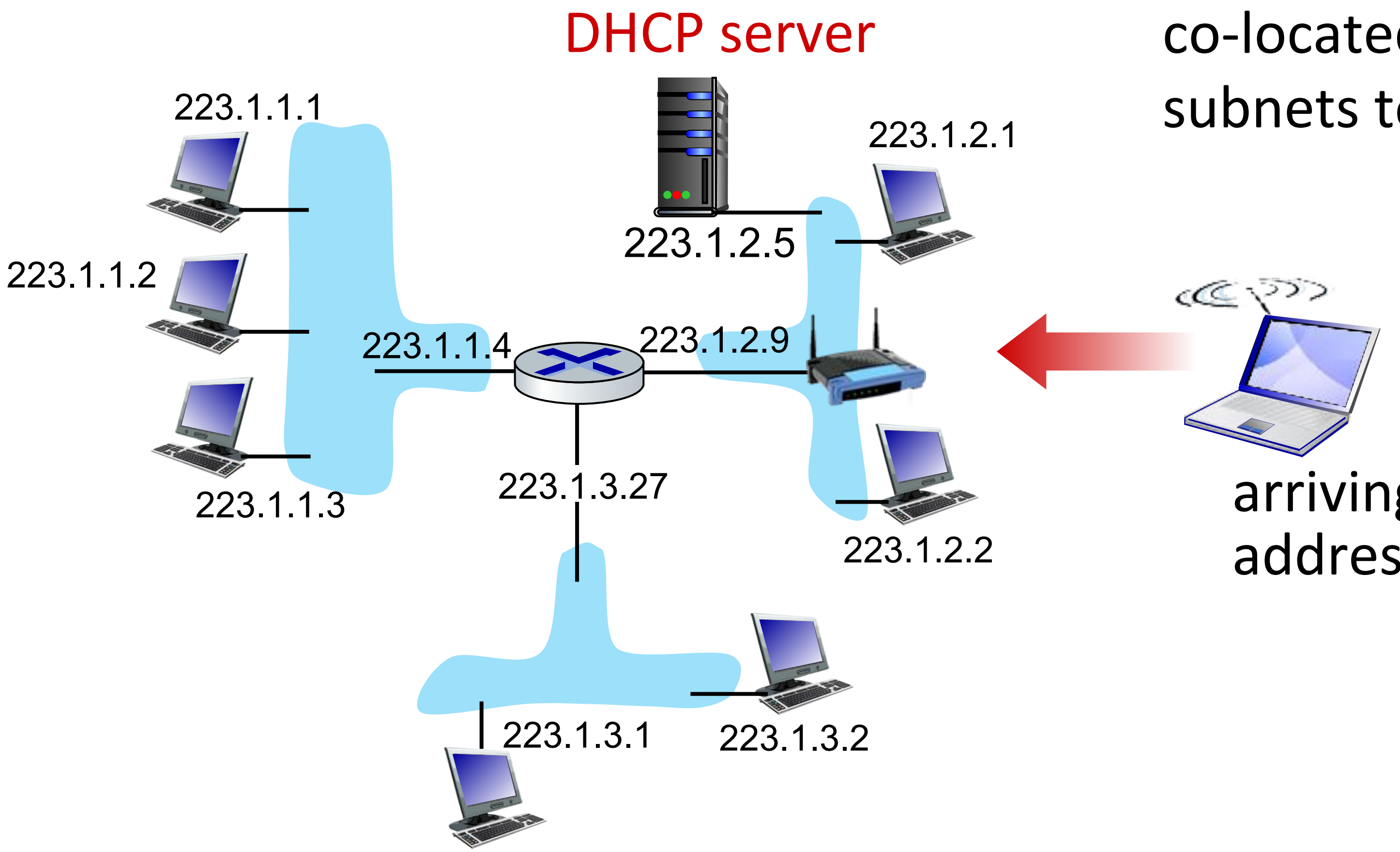
- can renew its lease on address in use
- allows reuse of addresses (host only hold address while connected/on)
- support for mobile users who join/leave network

DHCP overview:

- host broadcasts DHCP discover msg [optional]
- DHCP server responds with DHCP offer msg [optional]
- host requests IP address: DHCP request msg
- DHCP server sends address: DHCP ack msg

Work Flow: Discover → Offer → Request → ACK

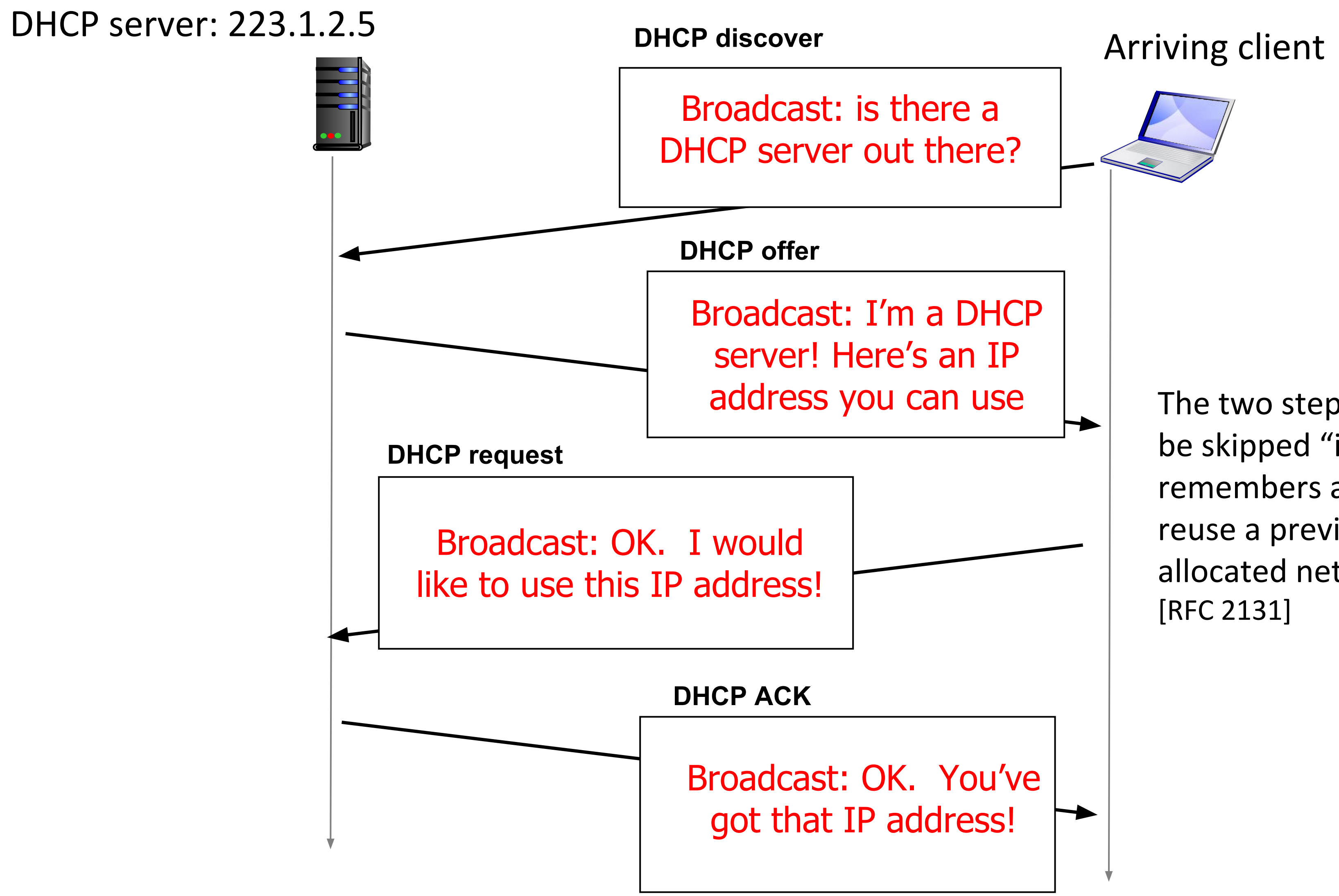
DHCP client-server scenario



Typically, DHCP server will be co-located in router, serving all subnets to which router is attached

arriving DHCP client needs address in this network

DHCP client-server scenario



The two steps above can be skipped "if a client remembers and wishes to reuse a previously allocated network address" [RFC 2131]

DHCP: more than IP addresses

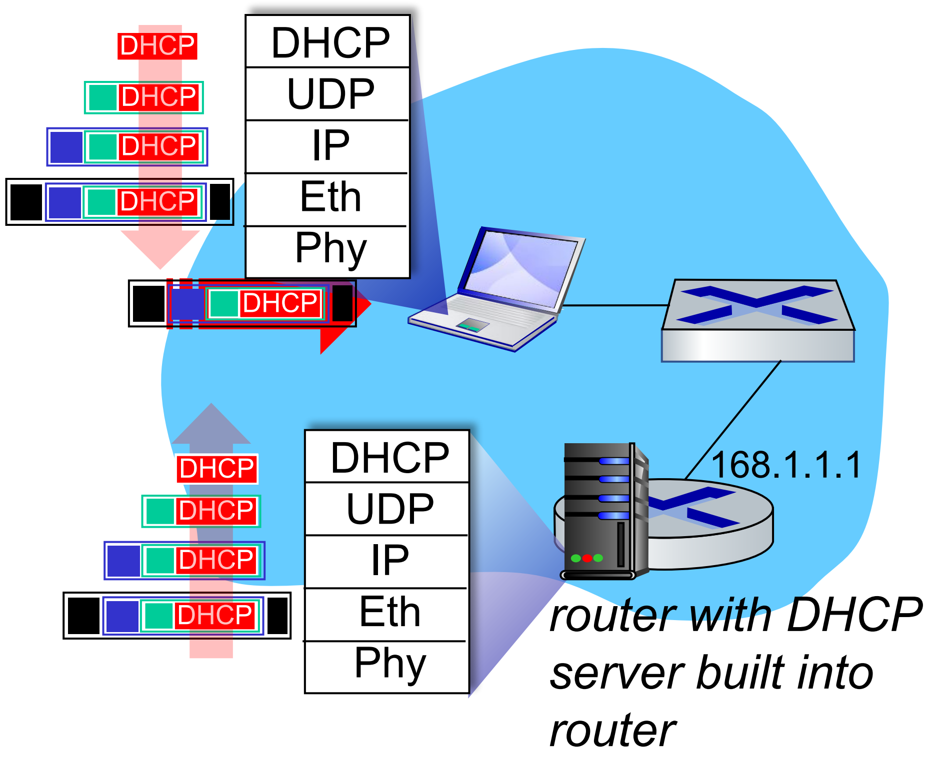
DHCP does **not only assign an IP address**, it also provides other important network parameters:

- address of first-hop router for client
- name and IP address of DNS sever:

→ Enables the host to translate domain names into IP addresses

- network mask (indicating network versus host portion of address)

DHCP: example



Ethernet → IP → UDP → DHCP

- Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server.
- A DHCP REQUEST message is encapsulated inside UDP, which is encapsulated in IP, which is finally encapsulated in an Ethernet frame.
  - The DHCP message is sent inside an Ethernet frame which broadcast (dest: FFFFFFFFFFFF) on LAN, received at router running DHCP server
  - The received Ethernet frame is DE multiplexed to IP, then to UDP, and finally delivered to the DHCP application.

IP addresses: how to get one?

Q: how does *network* get subnet part of IP address?

A: The subnet part of IP Address comes from the block of addresses allocated to the organization by the ISP

ISP's block	11001000 00010111 0001 0000	00000000	200.23.16.0/20

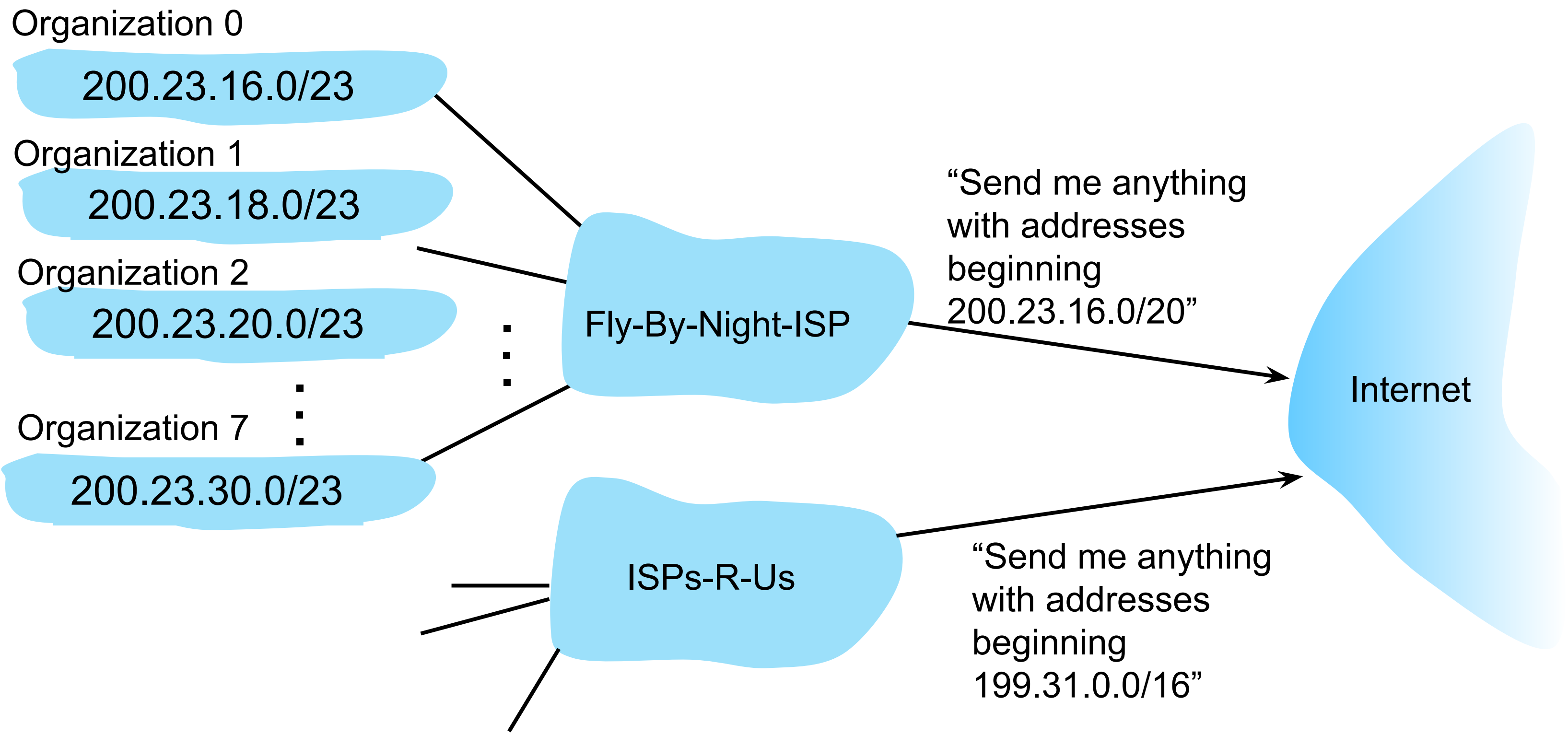
ISP has a large block: **/20** → first 20 bits are network bits. That gives the ISP 4096 addresses. Now The ISP wants to create 8 smaller blocks for 8 organizations.

So ISP needs to split 4096 addresses into 8 equal: parts: 4096÷8= 512

Organization 0	11001000 00010111 0001000 0	00000000	200.23.16.0/23
Organization 1	11001000 00010111 0001001 0	00000000	200.23.18.0/23
Organization 2	11001000 00010111 0001010 0	00000000	200.23.20.0/23
...	.....	....	....
Organization 7	11001000 00010111 0001111 0	00000000	200.23.30.0/23

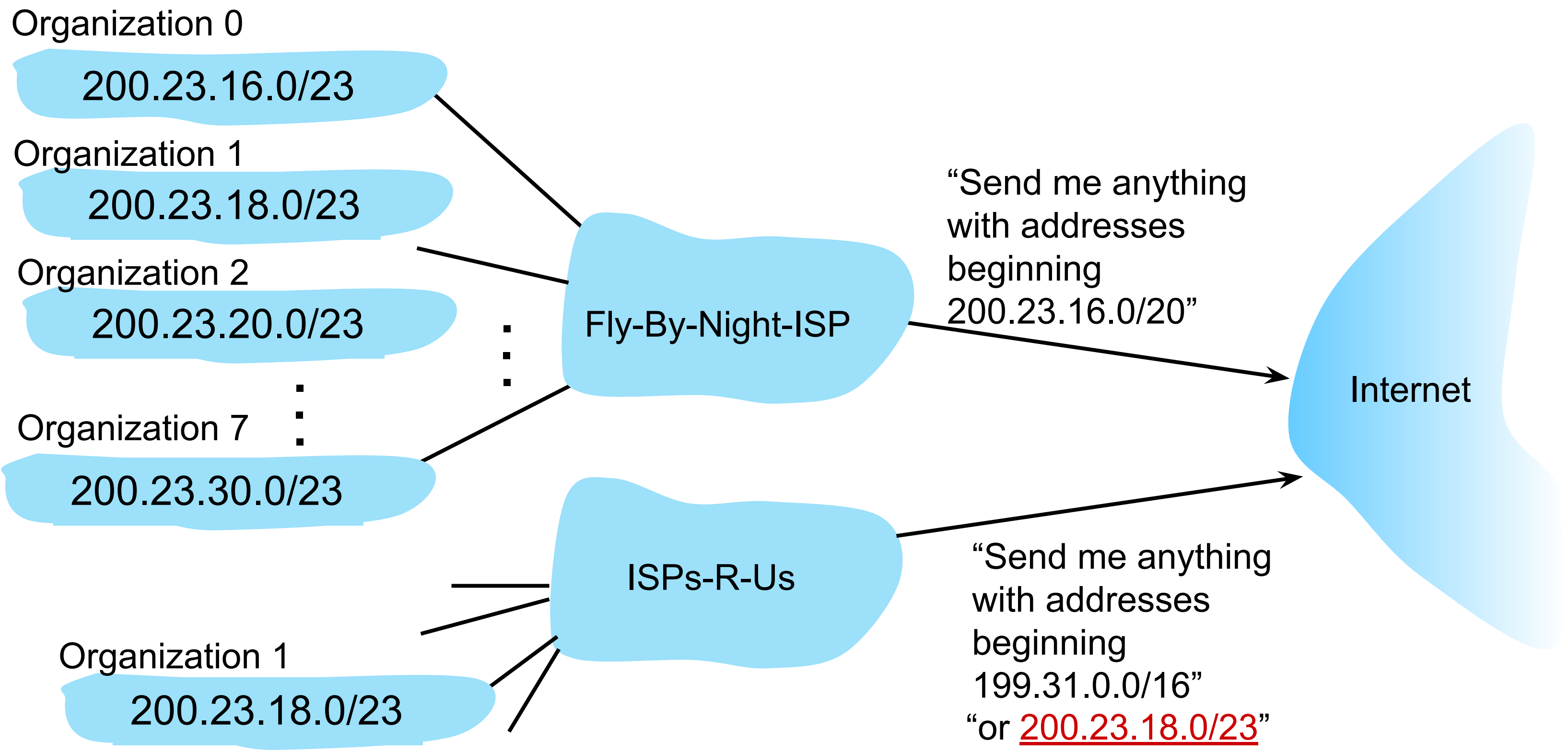
Hierarchical addressing: route aggregation

hierarchical addressing allows efficient advertisement of routing information:Fly-By-Night-ISP advertises a **general route** to the Internet: That means the Internet doesn’t need to know every individual organization; it only needs to know the ISP’s /20 block.



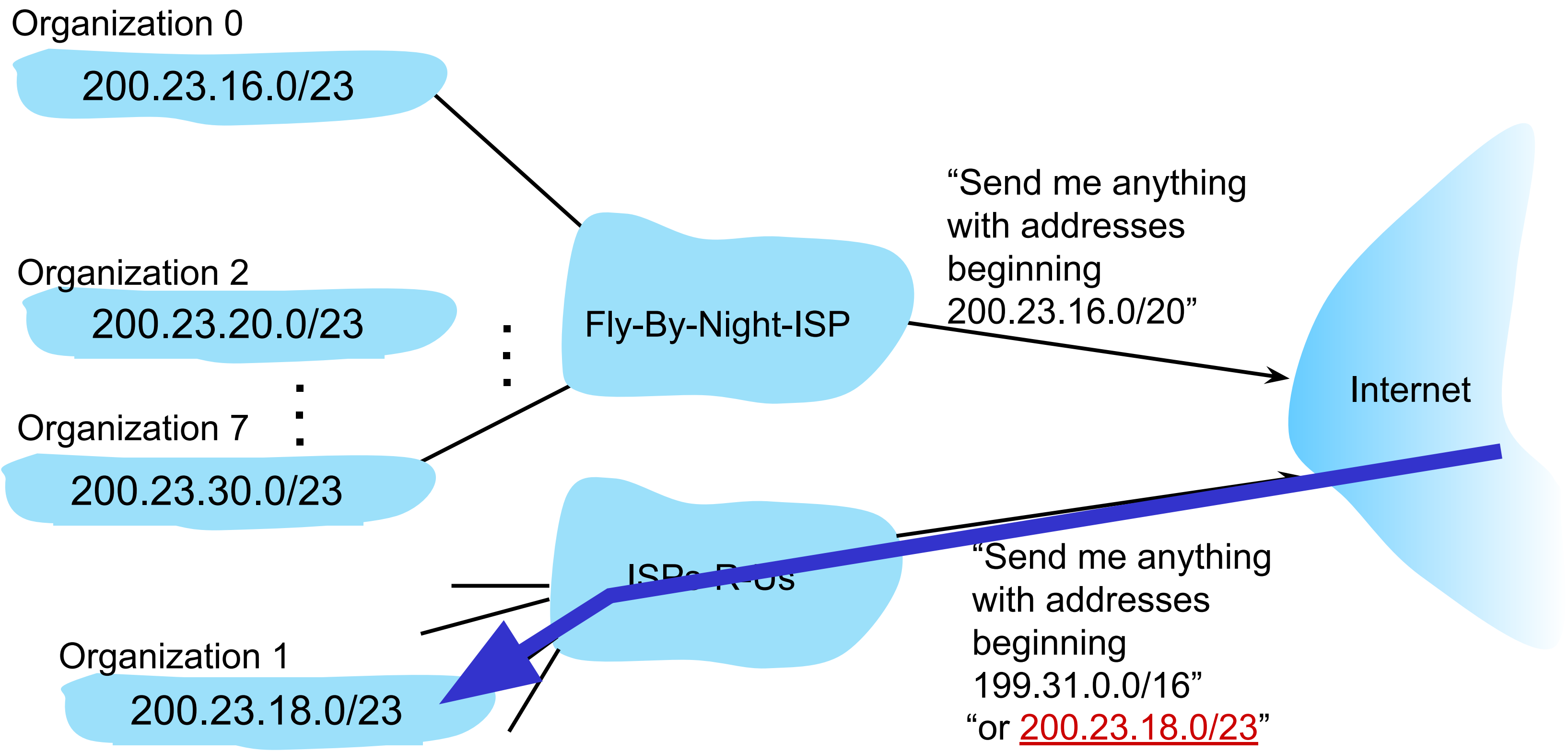
Hierarchical addressing: more specific routes

- Organization 1 moves to ISPs-R-Us. **ISPs-R-Us** now controls 200.23.18.0/23, which was part of the original ISP's /20 block. **ISPs-R-Us** advertises a more specific route for Organization 1:



Hierarchical addressing: more specific routes

- Organization 1 moves from Fly-By-Night-ISP to ISPs-R-Us
- ISPs-R-Us now advertises a more specific route to Organization 1



IP addressing: last words ...

Q: how does an ISP get block of addresses?

A: ICANN: Internet Corporation for Assigned Names and Numbers http://www.icann.org/

- allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
- manages DNS root zone, including delegation of individual TLD (.com, .edu , …) management

Q: are there enough 32-bit IP addresses?

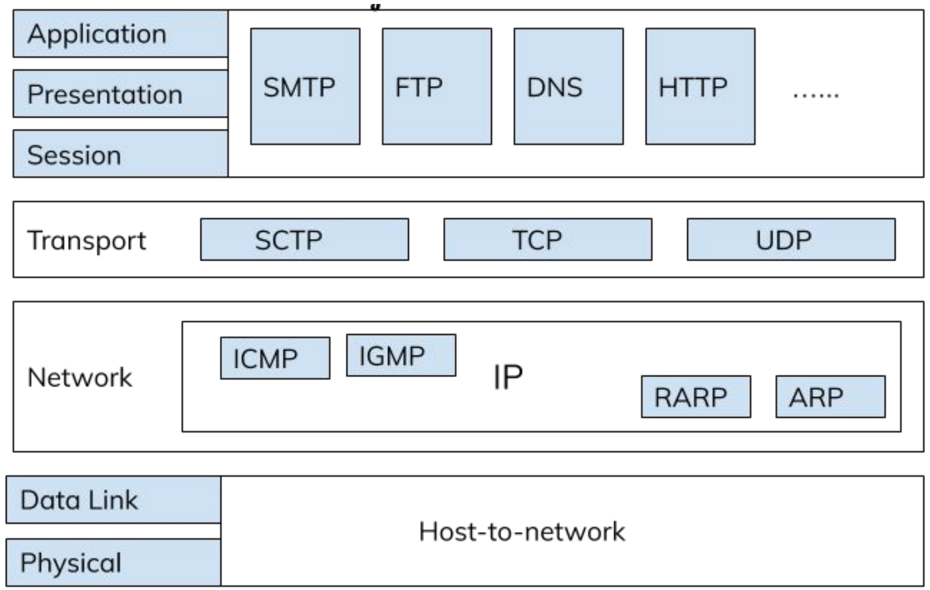
- ICANN allocated last chunk of IPv4 addresses to RRs in 2011
- NAT (next) helps IPv4 address space exhaustion
- IPv6 has 128-bit address space

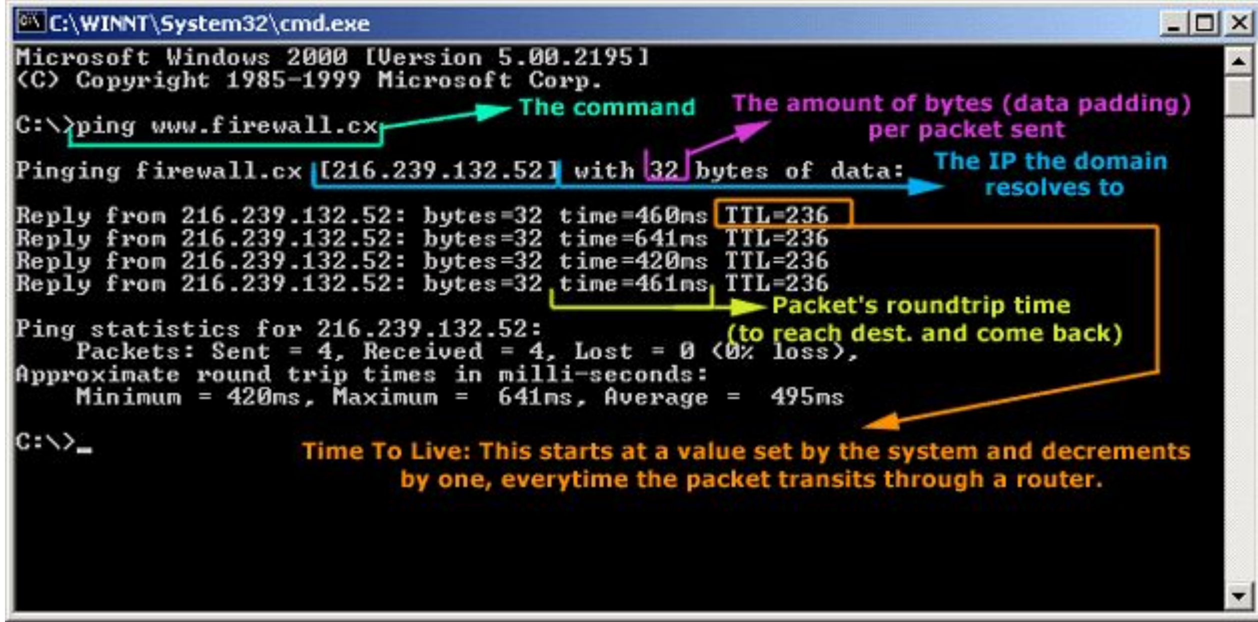
"Who the hell knew how much address space we needed?" Vint Cerf (reflecting on decision to make IPv4 address 32 bits long)

Internet Control Message Protocol (ICMP)

- ICMP is a network layer protocol used for error reporting and network diagnostics
- It works alongside IP, not as a data-carrying protocol
- ICMP messages are generated by routers and hosts
- Helps identify problems such as:
  - Destination unreachable
  - Packet time-to-live (TTL) expiration
  - Routing issues
- ICMP is essential for the proper functioning

and troubleshooting of IP networks





ICMP

- Echo Request / Echo Reply
  - Used by the ping command to test reachability
- Destination Unreachable
  - Indicates network, host, or port is unreachable
- Time Exceeded
  - Occurs when TTL expires (used by traceroute)
- Redirect Message
  - Suggests a better route to the sender
- ICMP improves network reliability but does not guarantee delivery

Category	Type	Message
Error-Reporting Messages	3	Destination unreachable
	4	Source quench
	11	Time Exceeded
	12	Parameter Problem
	5	Redirection
Query Message	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply
	17 or 18	Address mask request or reply
	10 or 9	Router Solicitation or advertisement

Lecture End